



Reading Resistance

An interpretable, honestly-benchmarked ML classifier for antibiotic resistance from bacterial genomes

Predict resistant vs. susceptible per antibiotic — validated on unseen lineages, benchmarked against a transparent gene-lookup, explaining the genes behind every call.

8 WHO-priority pathogens · 36 models · fully computational · public data · two-person team

THE TEAM

| Two people, complementary by design

Software / ML engineer

- Data pipeline & feature extraction
- Phylogeny-aware splitting
- Modeling, calibration, SHAP tooling
- The interactive demo

Microbiology M.Sc. \$

- Organism & drug selection
- Phenotype-label quality control
- Biological validation of determinants
- Clinical error trade-offs & limitations

Engineering builds the rigor; microbiology owns every biological decision and proves the model learned **real mechanisms**, not noise.

THE WHOLE SYSTEM IN ONE BREATH



Genome in → determinant features

Which known resistance genes & mutations
does this strain carry?



Per-drug R/S + the reason

Calibrated probability, and the genes that
drove the call.

Largest *internal* gain: for **cefoxitin**, a gene-lookup misses **96.3%** of resistant strains — resistance comes from **losing a porin**, not gaining a gene. Our model lifts AUC from **0.52 → 0.90**. **On external isolates this gap does not hold (slide 30) — the contribution we can defend is the measured generalization gap.**

PART 1

The problem & why it matters

Why genomes can predict resistance — and the two traps that fool naive projects.

WHY AMR

Resistance is written in the genome

- Bacteria resist by carrying a **gene** (an enzyme that destroys the drug)...
- ...or by a **point mutation** in the drug's target.
- Standard testing grows the strain against the drug — accurate but **slow (days)**.
- Whole-genome sequencing of clinical isolates is now routine → huge public collections of genomes **paired with lab phenotypes** already exist.

The opportunity

The raw material for genome→resistance prediction is sitting in public databases. The challenge isn't data — it's **honest evaluation**.

Two traps that inflate every naive result

1 · Clonal population structure

Bacteria are **clonal** — the data is full of near-twins from the same lineage. A **random** train/test split lets the model see a resistant strain in training and its near-identical cousin in test → it's rewarded for **recognizing the clone**, not understanding resistance. Every metric inflates.

2 · "It's just a database lookup"

If a model merely detects a known resistance gene, it's a **database query, not science**. The value is in what the model **adds** over that lookup.

Our whole design answers both: a **phylogeny-aware split** and an explicit **rules baseline**.

| What we set out to prove

Question. Can an interpretable model predict resistance from genomic features, **generalize to lineages it has never seen**, and be honestly benchmarked against a gene-lookup and published work?

Hypothesis. A gradient-boosted model on determinant features, under a phylogeny-aware split, will:

a · strong per-drug discrimination on unseen lineages

b · match/beat the gene-lookup, cutting dangerous errors

c · top features = established mechanisms

Nine non-negotiables, enforced in code

✓ Split by **lineage**, never randomly

✓ Always report the **gene-lookup baseline**

✓ **One model per antibiotic**

✓ **VME / ME** reported before accuracy

✓ **Determinant features only** (no k-mers)

✓ Always **calibrate** probabilities

✓ No **state-of-the-art** overclaim

✓ **Scope discipline**

✓ **Never fabricate** a genome/label/number

Written before touching data. A **test fails the build** if a single lineage crosses the split.

PART 2

Setting up from scratch

How anyone rebuilds this from nothing — the environment, tools, and reproducibility spine.

THE STACK

| Free, public, laptop-scale

Python data-science

Python 3.11 · pandas · numpy · scikit-learn ·
XGBoost / LightGBM · **SHAP** · matplotlib · scipy

Bioinformatics (bioconda)

AMRFinderPlus 4.2.7 — resistance-gene caller
(the feature engine)
mlst 2.33.1 — lineage typing

Optional extensions

Streamlit — the demo web app
PyTorch + **ESM-2** (fair-esm) — protein-
language-model MIC extension
Runs on a laptop; ESM-2 embeds on Apple-
Silicon GPU in minutes.

conda, not pip the bio tools ship via bioconda

INSTALL

From bare machine to a prediction in 4 commands

```
# 1 · Get the code
git clone https://github.com/kappaborg/amr-resistance-predictor
cd amr-resistance-predictor

# 2 · Build the exact pinned environment (conda – for the bio tools)
conda env create -f environment.yml
conda activate amr-resistance-predictor

# 3 · Prove it works – runs the zero-lineage-leakage test
make test          # -> 20 passed
```

`environment.yml` pins exact tool + DB versions, so anyone regenerates the **same features**.

| Nothing scientific is hard-coded or untracked

Fixed seed 42

everywhere — splits, models, bootstraps

config.yaml

organism, drugs, thresholds, model choice

data/manifest.md

every source, version, query, count

Makefile

one command per pipeline stage

decisions.md

53 logged decisions — the lab notebook

data/raw/ git-ignored

regenerates from the manifest

PART 3

The data

Public resources, data-driven organism choice, and the label discipline that keeps it honest.

RESOURCES

Five public sources, no wet lab

Role	Source	What it gives
Genomes + labels	BV-BRC	Assembled genomes + laboratory R/S phenotypes
Reference determinants	CARD	Catalog of known resistance genes + mutations
Feature annotation	AMRFinderPlus	Genome → gene/mutation presence-absence
Lineage typing	mlst	Each genome → sequence type (lineage)
Validation reference	Published AMR-ML	Honest external comparison

| Organism chosen by data, not gut

- Queried BV-BRC for **lab-phenotype counts** across candidate pathogens/drugs.
- Picked for **abundant, class-balanced** resistant/susceptible data.
- **Locked: Klebsiella pneumoniae** (taxon 573) + a 5-drug panel.

Why *K. pneumoniae* fits the thesis

Carbapenem resistance is exactly where single-gene rules **fail** — porin loss + β -lactamase combinations → real, catchable dangerous errors.

Strong clonal structure → the lineage-aware split genuinely **bites**.

5-drug panel = **5 distinct mechanisms**: meropenem · gentamicin · ciprofloxacin · TMP-SMX · ceftazidime.

LABEL DISCIPLINE

| Lab phenotypes only — avoiding circularity

Kept ✓

evidence = "Laboratory Method" — actual measured susceptibility.

Excluded ✗

Computational AMR predictions — training on them = predicting another model's output (circular).

MIC / SIR → binary **R/S**; the ambiguous **"Intermediate"** class is **dropped**, **not guessed** (⚕ needs sign-off).

ACQUISITION

Rigor shows in the bugs you catch



Caught: silent truncation

The genome API defaulted to **25 records** → assemblies truncated to ~2 of ~5.5 Mbp. A truncated genome **drops genes** → false-absent features → corrupted labels. Caught by a file-size integrity check; fixed and re-downloaded everything.

Robust by design

BV-BRC REST API — no bulk download, size flagged first. Downloader **retries with backoff** and **resumes** — one timeout shouldn't kill an overnight run.

Decision log entries #15—#16.

Literature-aligned thresholds (MIMAG)

- Completeness $\geq 90\%$, contamination $\leq 5\%$
(Bowers 2017)
- Contigs ≤ 500 · length 4.5–7.5 Mbp
(Kleborate range)
- Length cap widened to 7.5 Mbp — MDR
plasmid strains legitimately reach ~6.6
Mbp; a tight cap discards **real resistant**
strains.
- Drops only **1.9%**, and the drops are **class-**
balanced → no label bias.

3,850

QC-passed *K. pneumoniae* genomes

688

determinant features

PART 4

Methods — the science core

Features, the lineage split (the crown jewel), models, calibration, and the clinical metric.

| Genome → binary determinant matrix

- AMRFinderPlus --organism *Klebsiella_pneumoniae* on every genome.
- Assemble a binary matrix: rows = genomes, cols = determinants, **1 = present**.
- Keep only true **AMR** elements — exclude virulence / stress genes.
- Include **point mutations** (gyrA, parC...) — the whole story for fluoroquinolones.

3,850 × 688
genomes × determinants (*K. pneumoniae*)

Fast & interpretable — **not** black-box k-mers.

★ THE MOST IMPORTANT SLIDE

The lineage-aware split

- **mlst** assigns each genome its **sequence type** (lineage).
- StratifiedGroupKFold holds out **whole lineages** for testing — the model is tested only on sequence types it has **never seen**.
- R/S balance preserved across the split.
- Untypeable genomes each get a **unique** synthetic lineage — they can never secretly pool into one leaky group.

The guard

A **test asserts zero lineage overlap** between train and test — for **all 8 organisms**. If one sequence type ever appears in both, **the build fails**.

`tests/test_split.py` · 20 tests pass

This one choice is **why every number afterwards is trustworthy**.

| We quantified the leakage we avoided

Same model, scored under a naive **random** split vs our **lineage** split:

+0.010

mean AUC inflation from a random split
(max +0.032, cefoxitin)

The inflation is **small** — because our features encode **mechanism**, **not ancestry**. A genome resists because it carries blaKPC, not because of its clade.

Contrast

K-mer models — which secretly encode lineage — lose **0.1–0.2 AUC** on this same test.

Our small gap is itself **evidence we learned biology**. We still report the honest lineage number everywhere.

MODELING

| One model per antibiotic, three honest tiers

1 · Rules baseline

"Resistant if any known gene for this drug is present." The transparent **yardstick**.

2 · Logistic regression

Simple, linear, interpretable reference.

3 · XGBoost

Gradient-boosted — the primary model. Calibrated + thresholded.

Resistance is drug-specific → **never pool drugs into one label**. All three reported so "what ML adds" stays honest.

| The gene-lookup baseline is organism-aware

"Ampicillin resistance" is **not** the same gene in every bug:

E. coli → β -lactamases

Enterococcus → pbp5

S. pneumoniae penicillin → pbp1a/2b/2x mosaic

A single generic rule would be a **strawman** baseline in some organisms. ORG_RULES matches the **actual mechanism** per (organism, drug) — so when we beat the baseline, we beat a **real one**.

Probabilities that mean something · errors that matter

Calibration

Isotonic calibration + **Brier score** + reliability curves. A "90% resistant" must be right ~90% of the time — an uncalibrated 0.9 that's really 0.6 is dangerous.

VME / ME — reported first

Very-major error (VME): resistant called susceptible — patient gets a drug that won't work. **The dangerous miss.**

Major error (ME): susceptible called resistant — an unnecessary alarm.

Operating point tuned to cap **VME $\leq$ 3%** (CLSI/FDA-style).

Capping VME **raises** ME on hard drugs — a clinical trade-off, made **in the open**, never hidden.

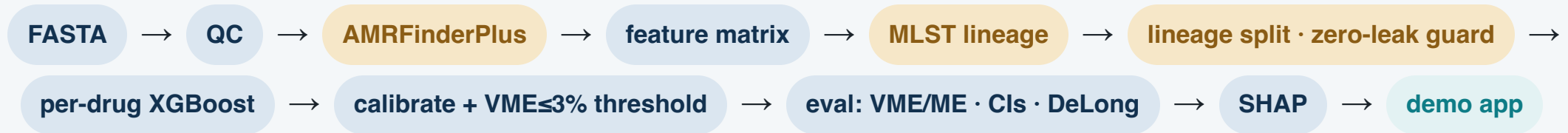
PART 5

Training & the pipeline

The assembly line — every box is a file you can run.

END-TO-END

Genome → explained, benchmarked prediction



```
# one command per stage – or the whole spine at once  
make data · features · split · train · eval · figures      # make all
```

Two decision gates: "enough R **and** S per drug?" and "generalizes to unseen lineages?" — if not, loop back to the split.

HOW A MODEL IS BUILT

No peeking at the test set

- Fit XGBoost on the **training lineages**.
- Calibrate (isotonic).
- Pick the $VME \leq 3\%$ threshold on **out-of-fold** training predictions — never the test set.
- Evaluate **once** on the held-out lineages.

Deployable bundle

model + calibrator + threshold + feature columns →
one `.joblib` per organism-drug

36

deployable models (8 organisms)

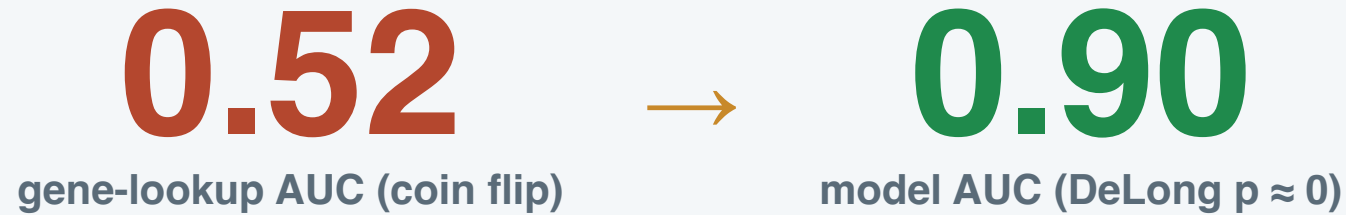
PART 6

Results

Strong discrimination on unseen lineages — and honest about where the lookup already wins.

THE HEADLINE

Cefoxitin — where the lookup fails



Resistance is driven by **losing the ompK36 porin** — the drug can't get in. A *gene-lookup* is blind to a **missing** feature (misses 96.3% of resistant strains). The model learns the absence pattern.

The cleanest *internal* demonstration of what ML adds over a database query — and, on external isolates, the drug that fails hardest (0.905 → 0.596). Both facts belong to the same slide.

Strong & significant across the board

Drug	AUC model	AUC rules	VME	ME	DeLong p
ciprofloxacin	0.983	0.911	2.3%	7.7%	7e−36
TMP-SMX	0.977	0.885	2.3%	18.3%	9e−51
meropenem	0.968	0.933	1.3%	55.9%	5e−24
gentamicin	0.981	0.948	1.6%	32.4%	1e−30
cefoxitin	0.905	0.518	0.7%	94.2%	~0

VME point estimates at/below the 3% bar on all five · beats the lookup significantly on all five. Honest caveat: lineage-clustered CIs are wide (meropenem 1.3% [0.3–3.5]) — target met *in expectation*, not guaranteed at the upper bound. High ME on meropenem/cefoxitin = the VME-first trade-off, shown with flags.

| Where we do **not** win — said first

Gentamicin

Rules already excellent (AUC 0.948) — direct enzyme signal. ML only **matches** it. We say so.

Salmonella chloramphenicol

Rules baseline **beats** our model. Reported plainly.

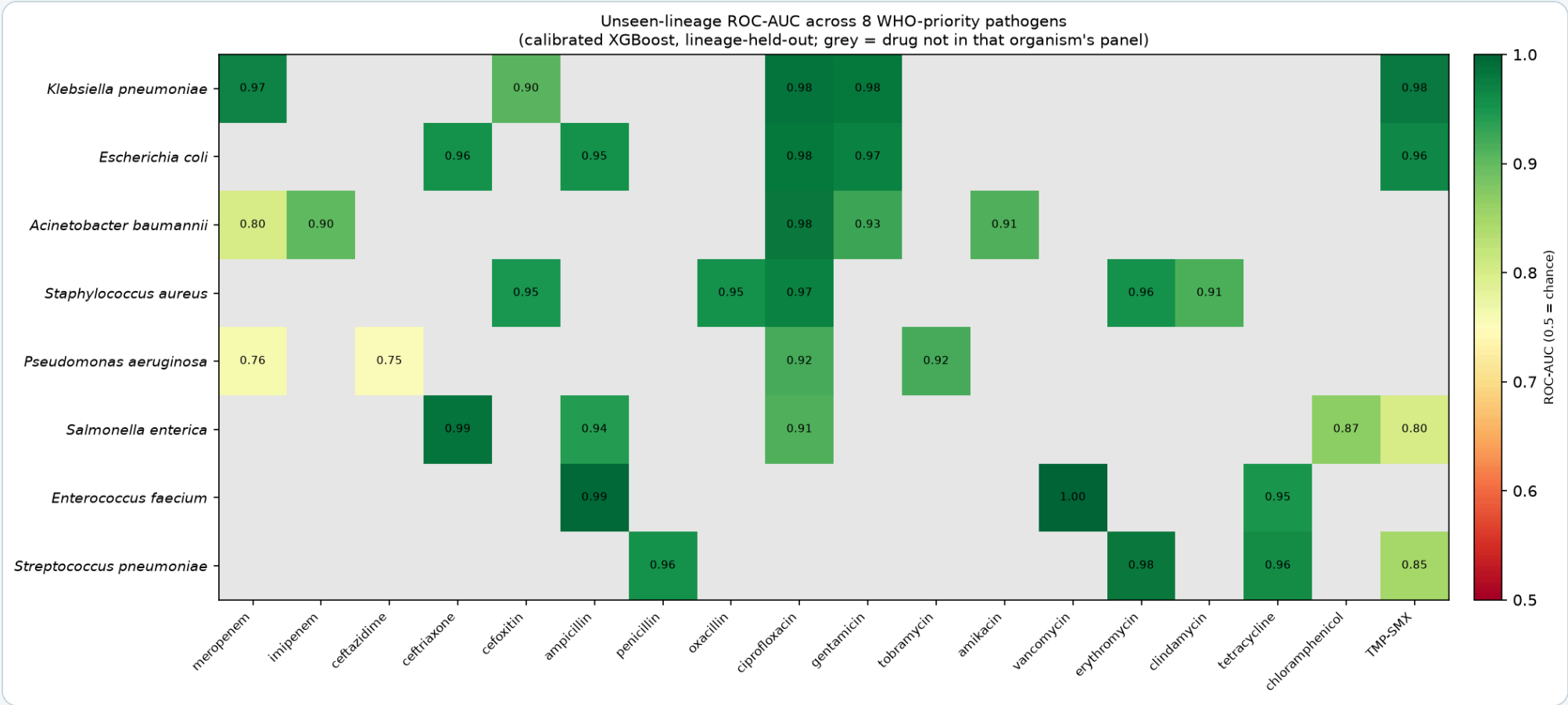
S. pneumoniae TMP-SMX

Causal folA/folP **not in the catalog** → any signal is **co-selection**, not mechanism. Labeled as such.

An award isn't won by claiming you win everything — it's won by **knowing exactly where you don't**.

GENERALIZATION

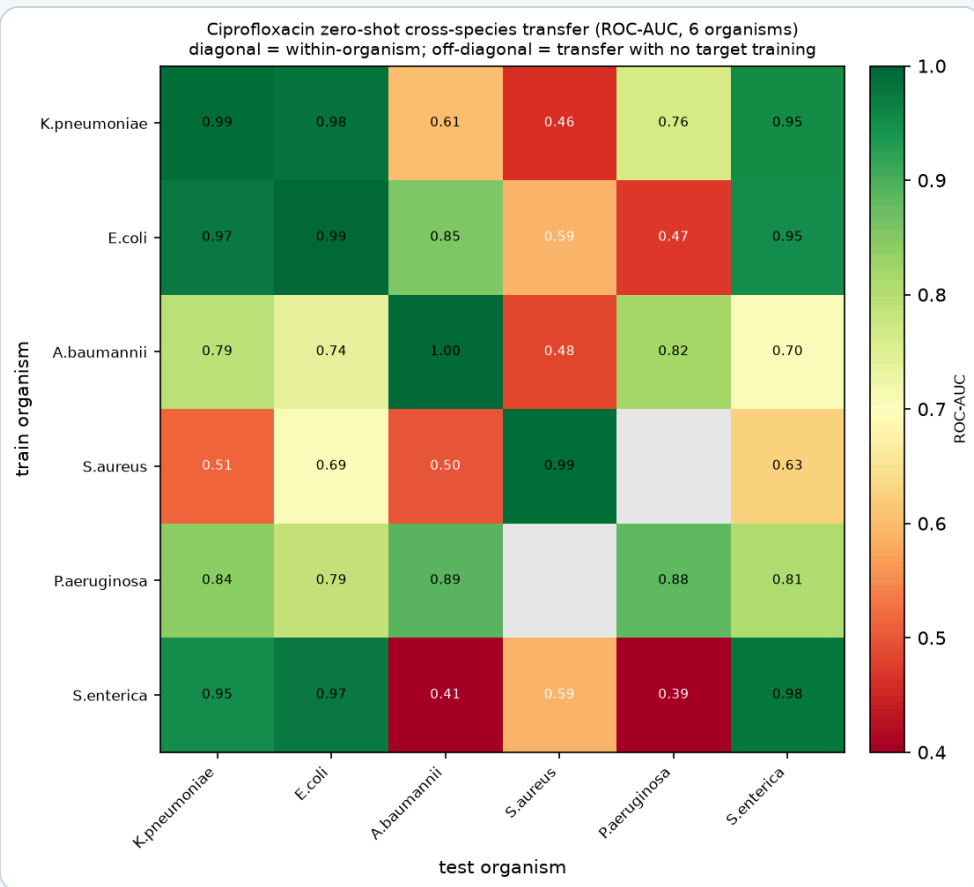
8 WHO-priority pathogens · one pipeline



Unseen-lineage ROC-AUC, 8 organisms × up to 18 drugs (36 calibrated models). ROC 0.84–0.998 (single split; 0.751–0.998 under pooled CV).

CROSS-SPECIES ZERO-SHOT TRANSFER

The model learned biology, not lineage



Train on one species, test zero-shot on another. Off-diagonal = transfer.

Ciprofloxacin transfers **across Gram-negatives** (shared gyrA/parC target + numbering)...

...and **collapses** to Gram-positive S. aureus (uses grlA, different gyrA numbering).

A mechanistically-**expected** boundary — impossible if the model had memorized lineage artifacts.

| The claim, backed by statistics

DeLong test

Model beats the gene-lookup **significantly** on discrimination — every drug, p from $2e-14$ to ~ 0 .

Lineage-clustered bootstrap

95% CIs that resample **whole lineages**, not genomes — honest for a small resistant class.

Imbalance-robust

MCC + PPV/NPV at realistic prevalence — accuracy alone would flatter us.

Our AUCs sit in/above the **published band** — but under a **stricter** lineage split, with interpretability + clinical errors most studies omit. Claim: **honestly measured**, not state-of-the-art.

| We ran it on someone else's isolates. It got worse.

Drug	Internal	External	ST never seen	vs gene-lookup
meropenem	0.968	0.795	0.685	+0.027
gentamicin	0.981	0.826	0.869	+0.030
ciprofloxacin	0.983	0.835	0.821	+0.065
TMP-SMX	0.977	0.806	0.774	+0.092
cefoxitin	0.905	0.596	0.592	+0.027 · rules win on ST-novel

1,143 isolates curated by other groups. Models frozen — no retraining, no re-thresholding, one pass. The analysis plan was registered before a single genome was downloaded, so this result could not be quietly dropped.

| Why it dropped — and what it proves

Not a broken pipeline ✓

Checked first, because a bug looks exactly like this.

External genomes carry **14.4** determinants vs **15.1** in training; **zero** came back empty. The drop is **real**.

Prevalence shifted

Cipro **-24 pts**, cefoxitin **-27 pts** resistant — partly *our* doing: we enriched training for resistance. Explains the over-calling, **not** the AUC drop.

Co-selection doesn't travel

Cefoxitin leaned most on **co-carried** MDR markers — and collapses. The **mechanistic** AmpC lookup barely moves (0.518→0.569).

A real mechanism travels. A population-specific correlation does not. **We wrote this down as a limitation before we ran the test** — the external cohort turned a caveat into a measurement.

PART 7

Advanced & novel extensions

MLC regression · protein language models · an honest negative ·
uncertainty & clinical utility.

BEYOND BINARY

MIC regression — predicting the *level*

Clinicians think in **MIC** — the minimum inhibitory concentration. We predict it continuously and score it the way regulators do: **Essential Agreement** (within ± 1 two-fold dilution).

Censor-aware scoring — most real MICs are ">32" / " ≤ 0.5 " bounds; naive clipping distorts the metric.

Drug (K. pneumoniae)	Essential Agreement
ciprofloxacin	92.7%
gentamicin	85.1%
cefoxitin	79.9%
meropenem	75.4%

ESM-2 — allele resolution for the hard carbapenems

Presence/absence discards **which allele** — but carbapenem MIC depends on it (KPC-2 vs KPC-3; OXA/NDM/VIM). We embed each resistance protein's **actual sequence** with ESM-2, so different alleles get different vectors.

Real, significant gains — exactly where predicted

K. pneumoniae **meropenem** EA **68 → 73%** (+4.8, p=0.0004)

A. baumannii **imipenem** EA **54 → 60%** (+6.7, p=0.0001)

Neutral where determinants already saturate (gentamicin, cipro). 650M gave no extra gain → 150M is the honest, laptop-scale choice.

| We implemented the SOTA graph model. It lost.

Isolate-similarity GNN

Each genome a node, edges by similarity (AMR-GNN style). On identical lineage-grouped folds it was **worse than a plain MLP on all 5 drugs.**

Why — and we can explain it

Our curated determinant features already capture the mechanistic signal cleanly. A population-structure graph only re-adds the **lineage information we deliberately avoid**, and its smoothing blurs sharp gene→phenotype boundaries.

"We tested state-of-the-art, it didn't win, and here's exactly why" — a **stronger** position than assuming it would help.

| A model that knows when it doesn't know

Conformal abstention

Per-class coverage + an explicit "**defer to phenotypic testing**" option. Confidently handles ~95% for well-powered drugs; for hard carbapenems, ~55% at **near-zero VME**, defers the rest. Validated under lineage shift.

Decision curve & risk-coverage

Net benefit beats the lookup + treat-all across the VME-averse region. Deferring the least-confident ~**30%** roughly **halves VME** (meropenem 8.7% → 3.5%).

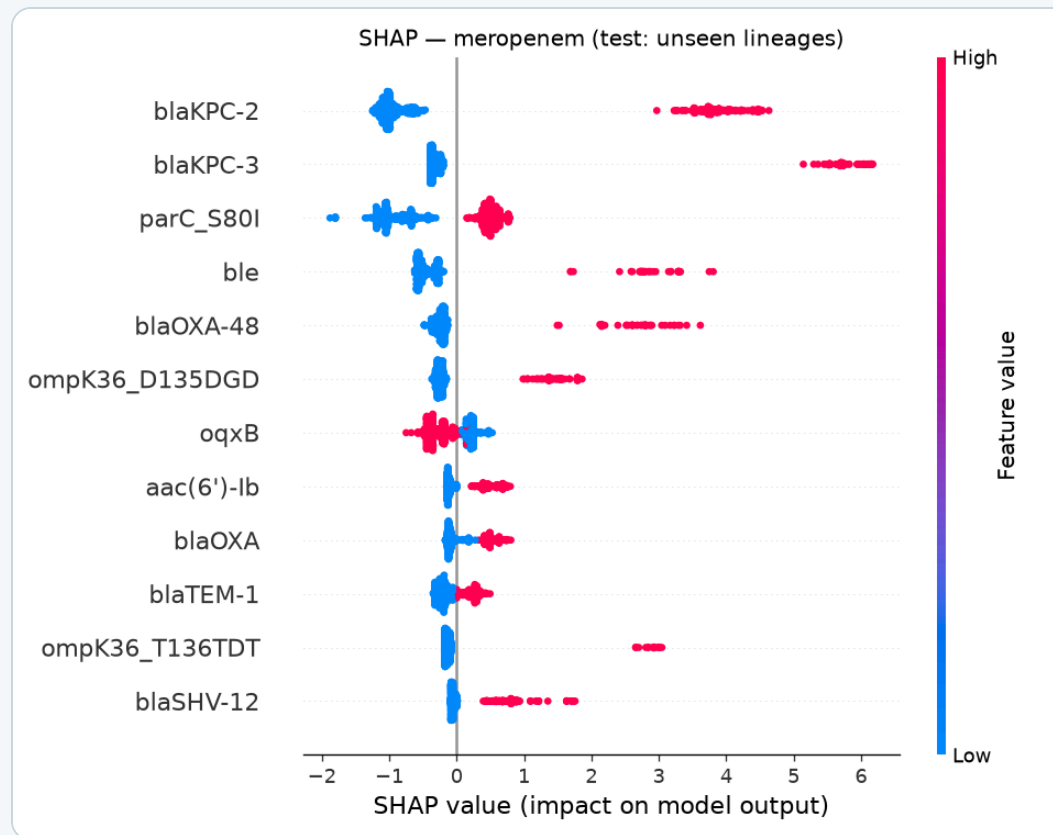
PART 8

Interpretability & biological validation

The microbiologist's section — does the model rely on real mechanisms?

SHAP

Every call names its genes



Meropenem SHAP — blaKPC-2 / blaKPC-3 rank as the top determinants (the carbapenemase). Per-strain, auditable.

Top features = textbook mechanisms

Drug	Model's top determinants	Known mechanism	✓
ciprofloxacin	gyrA, parC (QRDR) + qnr	DNA gyrase / topoisomerase target mutation	✓
gentamicin	aac(3) acetyltransferases	Aminoglycoside-modifying enzyme	✓
TMP-SMX	sul1, dfrA	DHPS / DHFR bypass	✓
meropenem	blaKPC-2, blaKPC-3	Carbapenemase	✓
cefoxitin	ompK36 porin loss	Reduced drug entry (loss, not gain)	✓
S. aureus oxacillin	mecA	Altered PBP2a	✓
E. faecium ampicillin	pbp5	Altered penicillin-binding protein	✓
S. pneumoniae penicillin	pbp1a/2b/2x mosaic	Altered PBPs (mosaic)	✓

Honest caveat: co-selection inflates some co-carried plasmid markers → read per-strain SHAP, not just averages. Catalogued determinants only → no novel-gene discovery (needs pan-genome features).

PART 9

The demo

A working web app — genome in, explained per-drug calls out.

8 organisms · upload a genome · get explained calls

- Pick an organism → cached genome or **upload a FASTA**.
- Runs AMRFinderPlus live → per-drug **R/S** + calibrated P(resistant) + SHAP **drivers**.
- Optional **AI clinical narrative** — explanation layer only, **never** changes a call; deterministic template with no API key.

```
# live  
python -m streamlit run src/app/streamlit_app.py
```

Verified example

S. aureus genome →

oxacillin RESISTANT (P=0.94)

drivers: **mecA** ↑, erm(C) ↑

ciprofloxacin RESISTANT (P=0.98)

drivers: **gyrA_S84L**, **parC_S80F** ↑

SAFETY

| The species guard

The model is a function of the determinant vector — it will "predict" on **anything** (a fungal file once produced a spurious call).

So the app verifies species **two independent ways**:

- **MLST scheme** (species-specific typing)
- **Intrinsic marker genes** (works on protein input too)

Mismatch → predictions **withheld**.

Every screen

"Research / decision-support only — **not a diagnostic**. Complements, does not replace, phenotypic susceptibility testing."

PART 10

Positives & negatives — honestly

Both belong on the slide. The limitations are where the honesty thesis pays off.

STRENGTHS

| What makes this award-grade

- **Honest by construction** — lineage split enforced by a build-failing test.
- **Beats the lookup where it matters**, with DeLong significance.
- **Generalizes** across 8 pathogens + the Gram divide; transfers along mechanistic lines.
- **Interpretable** — every call named & biologically validated.
- **Clinically framed** — VME-first, calibrated, abstention, net-benefit.
- **Beyond binary** — MIC + ESM-2 allele resolution.
- **Rigorous negatives** — GNN tested & rejected with a reason.
- **Reproducible** — pinned env, seeds, manifest, 53 decisions, 20 tests.

| The questions a judge would ask — asked first

- **Scope** — 8 organisms, ~3–5 drugs each; not a universal tool.
- **Co-selection** inflates some co-carried determinants (reduced, not gone).
- **High ME at VME $\leq$ 3%** for hard drugs — a threshold trade-off, not a discrimination failure.
- **Database bias** — CARD/BV-BRC over-represent well-sequenced regions.
- **Determinant features only** — no novel-gene discovery.
- **Honest weak spots** — Salmonella chloramphenicol, A. baumannii carbapenems, S. pneumoniae TMP-SMX.
- **A. baumannii meropenem MIC gain borderline** (p=0.051, under-powered).
- **Published comparison is band-level**, not a re-run head-to-head.

A limitation you raise yourself reads as **mastery**; dragged out of you, it reads as a gap.

Built to be checked, not trusted

Ethics

- Decision-support, **not a diagnostic device**
- Complements phenotypic testing — hence VME-first
- Public, de-identified data only; database bias stated
- Interpretability = an ethical safeguard

Reproducibility

- Pinned environment + fixed seed
- Data manifest + 53-entry decision log
- 20-test suite anyone can run
- Raw data regenerates from the manifest



Honestly measured

not "state-of-the-art" — trustworthy

1,143

external isolates · frozen models · pre-registered

8 · 36

pathogens · deployable models

The single most important thing we did: the lineage-aware split, enforced by a test. Everything downstream is trustworthy because of it.